

Power Formula Triangle

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS The triangle links power (P), work (W), and time (t). Cover the unknown to read the formula: $P = W \div t$, $W = P \times t$, and $t = W \div P$. Power is measured in watts (W), work (W) in joules (J), and time (t) in seconds (s).

Round 1 — Retrieve It

1. What three quantities does the power triangle connect?

2. Write the formula for power.

3. Write the formula for work (W).

4. Which variable do you cover to find time, and what formula results?

Round 2 — Sketch It

Draw the power triangle. Put power on top; put work (W) and time on the bottom. Circle the two quantities you would divide to find power.

Power — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. A motor does 100 J of work (W) in 5 s. What is its power?

2. A machine runs at 15 watts (W) for 4 s. How much work (W) does it do?

3. A pump does 240 J of work (W) at a power of 8 watts (W). How long does it take?

Round 4 — Reason from Evidence

Two machines each do the same 200 J of work (W). Machine A takes 10 s; Machine B takes 25 s. Use the triangle to find each machine's power, then say which machine is more powerful and how you know.

Power — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A

Articulate

Explain how the power triangle helps you find a missing quantity. Use one example that is not printed on the cue card.

C

Connect

Connect power to the idea that power is the rate of doing work — the work (W) done per unit of time. How does that meaning show up in $P = W \div t$?

E

Extend

Create your own problem that asks for time. Give the numbers and show the work using $t = W \div P$.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Power

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Power, work (W), and time.
2. $P = W \div t$.
3. $W = P \times t$.
4. Cover t; $t = W \div P$.

Sketch It — look for

Power on top, with work (W) and time on the bottom. Power is found by dividing work (W) ÷ time (the two bottom quantities).

Apply It

1. $P = W \div t = 100 \div 5 = \mathbf{20 \text{ watts (W)}}$.
2. $W = P \times t = 15 \times 4 = \mathbf{60 \text{ J}}$.
3. $t = W \div P = 240 \div 8 = \mathbf{30 \text{ s}}$.

Reason from Evidence — look for

Machine A: $P = 200 \div 10 = 20 \text{ watts (W)}$. Machine B: $P = 200 \div 25 = 8 \text{ watts (W)}$. Machine A is more powerful because it does the same work (W) in less time.

ACE — Extend look-for

Many answers possible. Check that the problem asks for time and the work correctly uses $t = W \div P$.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.